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Adam Gordon-Fennell, Ph.D.
Postdoctoral Research Associate
University of Washington, Seattle, WA 98195
agg2248@uw.edu

Dear Members of the Search Committee,

I am writing to apply for the tenure-track faculty position in the Rutgers Addiction Research Center and Brain Health Institute. I am currently a postdoc in Dr. Garret Stuber's lab at the University of Washington (UW), and I earned my Ph.D. in neuroscience in Dr. Michela Marinelli's lab at the University of Texas Austin.

Vision: Substance use disorder (SUD) is characterized by the transition from positive reinforcement to negative reinforcement throughout disease progression, but the neural circuits that motivate behavior under these two contexts remains poorly understood. My research goal is to uncover and separate neuronal circuits that motivate positive and negative reinforcement and then to use this framework to determine how these mechanisms are recruited to motivate drug taking throughout the development of SUD. To this end, my lab will first study how the lateral hypothalamus regulates dopamine release to motivate positive, negative, and opioid reinforcement. I will determine the dynamics of lateral hypothalamic neurons longitudinally using 2-photon imaging and the online relationship between lateral hypothalamic activity and brain-wide dopamine release using multi-site fiber photometry during behavior. Further experiments will focus on identifying targets for selective suppression of positive and negative reinforcement with the goal of informing future circuit-based therapies for SUD.

Prior Research and Skills: My technical and conceptual background makes me uniquely positioned to lead a lab to carry out the research vision outlined here. During my PhD and post-doctoral work, I studied subcortical neuronal circuits of motivated behaviors, including cocaine motivation and seeking behaviors. During my postdoc, to harness the power of 2-photon imaging to longitudinally track neurons throughout the development of disease states and across homeostatic demand, I designed and extensively validated a behavioral system (OHRBETS) for conducting a wide range of reinforcement and consummatory tasks in head-fixed mice. This system was published in *eLife* and is in use in at least 5 labs. I am now using this system to determine the role of hypothalamic regulation of striatal dopamine release during consumption and am preparing to submit this work. I am trained in conducting 2-photon imaging and the computational analysis of neuronal encoding of behavior. I was trained in head-fixed intravenous self-administration of heroin by Dr. Jim Otis's lab at MUSC.

Grants: I am confident that I will successfully obtain funding for my lab, as I obtained independent and training-grant funding throughout my scientific career including a K99/R00, NIDA F32, NSF GRFP, NIDA T32, and NIAAA T32. During my postdoc I also obtained a P30 pilot grant to establish head-fixed intravenous self-administration.

Mentorship and Service: I pride myself on creating an inclusive environment with mentorship that approaches each student as an individual and have been a mentor for 7 Ph.D. rotation students, 6 research technicians, and 9 undergraduate researchers. I am an active member of the UW neuroscience community as I lead presentations on data management, analysis, and visualization using programming languages; workshops on head-fixed behavioral approaches; and guest lectures in undergraduate and graduate neuroscience classes.

Collaboration: I have been actively engaged in multiple fruitful collaborations throughout my postdoctoral position related to the use and adaption of OHRBETS including labs of Drs. Michael Bruchas, Mitchell Roitman, Larry Zweifel, Marta Soden, Ajay Dhaka, James McCutcheon, and Ishmail Abdus-Saboor. From these projects I expect to see several co-authored manuscripts (see CV for more details) including one under review at *Cell*. In my own lab, I will continue to establish collaborative projects, especially related to designing head-fixed behavioral procedures to address novel research questions using cutting-edge imaging technologies.

Sincerely,

Adam Gordon-Fennell, Ph.D.